

MIDTERM ASSIGNMENT

ADVANCED STATISTICAL MECHANICS

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1. Use the fact that the entropy $S(N, V, E)$ of a thermodynamic system is an extensive quantity and thus a homogeneous function of degree 1 in its variables to show that

$$N \left(\frac{\partial S}{\partial N} \right)_{V, E} + V \left(\frac{\partial S}{\partial V} \right)_{N, E} + E \left(\frac{\partial S}{\partial E} \right)_{N, V} = S, \text{ leading to } N\mu = E + PV - TS = G.$$

Goal : Find expressions for S & G using the extensive property of Entropy

Solution

An extensive property f in variable x means it scales linearly with its variable.

$$f(\lambda x) = \lambda f(x)$$

Therefore, we can write $S(N, V, E)$ (which is extensive in N, V, E) as

$$S(N, V, E) = aN + bV + cE$$

Taking partial derivatives in N, V, E individually yields :

$$a = \left(\frac{\partial S}{\partial N} \right)_{V, E} \quad b = \left(\frac{\partial S}{\partial V} \right)_{N, E} \quad c = \left(\frac{\partial S}{\partial E} \right)_{N, V}$$

$$\text{Thus, } S(N, V, E) = \left(\frac{\partial S}{\partial N} \right)_{V, E} N + \left(\frac{\partial S}{\partial V} \right)_{N, E} V + \left(\frac{\partial S}{\partial E} \right)_{N, V} E$$

$$S = -\frac{\mu}{T} N + \frac{P}{T} V + \frac{1}{T} E$$

Rearranging

$$\mu N = E - TS + PV = G$$

2. Derive (i) an asymptotic expression for the number of ways in which a given energy E can be distributed among a set of N one-dimensional harmonic oscillators, with energy eigenvalues $\epsilon_n = \left(n + \frac{1}{2}\right) \hbar\omega$, $n = 0, 1, 2 \dots$, and (ii) the corresponding expression for the "volume" of the relevant region of the phase space of the system. Establish the correspondence between the two results, showing that the conversion factor ω_0 is precisely h^N .

Goal : Find the number of ways to distribute M quanta energy levels across N oscillators with total energy E .

Solution

The total energy E for N oscillators :

$$E = \sum_{i=1}^N \epsilon_n = \sum_{i=1}^N \left(n_i + \frac{1}{2}\right) \hbar\omega$$

Let M be the total quanta of the N particles such that $M = \sum_{i=1}^N n_i$

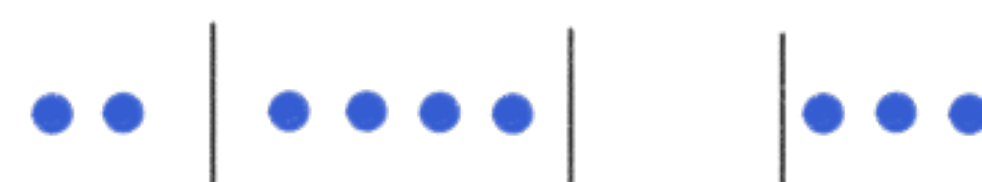
Then,

$$E = \hbar\omega \left[\sum_{i=1}^N n_i + \frac{1}{2} N \right]$$

$$\Rightarrow M = \frac{E}{\hbar\omega} - \frac{N}{2}$$

The number of ways to distribute M objects across N groups :

$$\Omega(E) = \binom{M+N-1}{N-1} = \frac{(M+N-1)!}{M! (N-1)!}$$



Permutation : M quanta
 $N-1$ separators
 $(M+N-1)!$

However, both quanta and separators are indistinguishable

$\therefore$ We divide by permutation of
 $M!$ quanta $(N-1)!$ separators

The textbook has a lovely asymptotic form in 3.8.25a.

$$\frac{(M+N-1)!}{M! (N-1)!} = \frac{\overbrace{(M+N-1)(M+N-2) \dots (M+1)}^{N-1 \text{ terms}}}{(N-1)!} \approx \frac{M^{N-1}}{(N-1)!} \text{ for } M \gg N.$$

For $M = \frac{E}{\hbar\omega} - \frac{N}{2} \approx \frac{E}{\hbar\omega}$ for $M \gg N$,

$$\Omega(E) \approx \left(\frac{E}{\hbar\omega} \right)^{N-1} \frac{1}{(N-1)!}$$

To get the multiplicity $\Gamma(E, \Delta)$ for a specific energy level $[E, E + \Delta]$,

we multiply by the number of eigenstates allowed in the energy interval $\frac{\Delta}{\hbar\omega}$ (2.5.15)

$$\Gamma_{\text{quantum}}(E, \Delta) \approx \Omega(E) \Delta E \approx \left(\frac{E}{\hbar\omega}\right)^{N-1} \frac{1}{(N-1)!} \cdot \frac{\Delta}{\hbar\omega} = \frac{E^{N-1}}{(\hbar\omega)^N} \Delta$$

Goal: Find the phase space volume $\Omega(E)$ for N 1D harmonic oscillators and use it to find the density of states $\omega(E)$.

Solution

The phase space volume for energy $\leq E_{\text{total}}$ is

$$\Omega(E) = \iint_{H \leq E} \prod_{i=1}^N dq_i dp_i$$

Let's explore the region of integration:

$$H(p, q) = \sum_{i=1}^N \frac{p_i^2}{2m} + \frac{1}{2} m \omega^2 q_i^2 \leq E$$

This looks pretty similar to a hypersphere but we have pesky coefficients.

An appropriate transformation from $u_i^2 = \frac{p_i^2}{2m}$ and $v_i^2 = \frac{1}{2} m \omega^2 q_i^2$ yields:

$$\sum_{i=1}^N u_i^2 + v_i^2 \leq E \rightarrow \text{This is a } 2N \text{ hypersphere with radius } E!$$

$$\text{Using the appropriate Jacobian: } \prod_{i=1}^N dq_i dp_i = \prod_{i=1}^N dv_i \cdot \sqrt{\frac{2}{m\omega^2}} dv_i \cdot \sqrt{2m} = \left(\frac{2}{\omega}\right)^N \prod_{i=1}^N dv_i du_i$$

Thus,

$$\Omega(E) = \iint_{H \leq E} \prod_{i=1}^N dq_i dp_i$$

$$= \iint_{\sum_i u_i^2 + v_i^2 \leq E} \left(\frac{2}{\omega}\right)^N du_i dv_i = \left(\frac{2}{\omega}\right)^N \frac{\pi^N}{N!} E^N$$

$$(7a) \text{ APP C } V_n(R) = \frac{\pi^{n/2}}{(n/2)!} R^n$$

Now, we can find the number of states in a specific volume $[E, E+\Delta]$ to get,

$$\begin{aligned}\Gamma_{\text{classical}}(E, \Delta) &= \frac{\partial \Omega(E)}{\partial E} \cdot \Delta \\ &= \left(\frac{2\pi}{\omega}\right)^N \frac{NE^{N-1}}{N!} \cdot \Delta \\ &= \left(\frac{2\pi}{\omega}\right)^N \frac{E^{N-1}}{N!} \Delta\end{aligned}$$

Great! This expression is similar to our result we found quantumly in (i)

This is exactly what we saw in 2.3.5!

$$\Gamma_{\text{quantum}} = \frac{\Gamma_{\text{classical}}}{\omega_0} \quad \text{where } \omega_0 \text{ is the fundamental volume of 1 microstate}$$

$$\omega_0 = \frac{\Gamma_{\text{classical}}}{\Gamma_{\text{quantum}}} = \frac{(2\pi)^N}{1/(\hbar)^N} = h^N \quad \square$$

We can see that the classical result still yields the correct result so long as we make use of the ω_0 factor.

3. A classical anharmonic one-dimensional oscillator, can be described by the following potential energy: $V(q) = aq^2 - bq^3 - cq^4$, where a, b, c are positive constants. The constant b and c are quite generally very small and $a = \frac{1}{2}m\omega^2$.

Using the partition function, $Z = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{-\beta(\frac{p^2}{2m} + V(q))} \frac{dp dq}{h}$, show that the leading contribution to the heat capacity of the oscillator is given by

$$C = \frac{3}{2}k^2 \left[\frac{c}{a^2} + 5 \frac{b^2}{a^3} \right] T,$$

And that to same order, the average value of the position coordinate x is given by

$$\langle q \rangle = \frac{3}{4} \frac{bkT}{a^2}.$$

Goal : Find the heat capacity C for an anharmonic oscillator $V(q) = aq^2 - bq^3 - cq^4$.

Solution

The Hamiltonian H for the 1D anharmonic oscillator is :-

$$\begin{aligned} H(p, q) &= K(p) + V(q) \\ &= \frac{p^2}{2m} + aq^2 - bq^3 - cq^4 \end{aligned}$$

The partition function for this classical system is :

$$\begin{aligned} Z(\beta) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp \left\{ -\beta \left(\frac{p^2}{2m} + aq^2 - bq^3 - cq^4 \right) \right\} \frac{dp dq}{h} \\ &= \int_{-\infty}^{\infty} e^{-\beta p^2/2m} dp \underbrace{\int_{-\infty}^{\infty} \exp \{ aq^2 - bq^3 - cq^4 \} dq}_I \cdot \frac{1}{h} \end{aligned}$$

Integral I is quite difficult to solve analytically, if possible. However, we can use our assumption given in the question that $b, c \ll 1$ and use a Taylor Expansion approximation of the cubic & quartic terms

$$\exp \{ -\beta a q^2 \} \cdot \exp \{ \beta (bq^3 + cq^4) \} = e^{-\beta a q^2} \left[1 + \beta (bq^3 + cq^4) + \frac{\beta^2}{2} (bq^3 + cq^4)^2 + \dots \right] \quad \text{Taylor Expansion of } e^{ax}$$

Returning back to our partition function.

$$Z(\beta) \approx \int_{-\infty}^{\infty} e^{-\beta p^2/2m} dp \int_{-\infty}^{\infty} e^{-\beta a q^2} \left[1 + \beta (bq^3 + cq^4) + \frac{\beta^2}{2} (bq^3 + cq^4)^2 \right] dq \cdot \frac{1}{h}$$

$$\text{We now use the result } \int_{-\infty}^{\infty} x^n e^{-ax^2} dx = \begin{cases} \sqrt{\frac{\pi}{2^n a^{n+1}}} (n-1)!! & \text{if } n \text{ even} \\ 0 & \text{if } n \text{ odd} \end{cases} \quad (A.1)$$

n	$\int_{-\infty}^{\infty} x^n e^{-ax^2} dx$
0	$\sqrt{\frac{\pi}{a}}$
4	$\frac{3}{2^2} \sqrt{\frac{\pi}{a^5}}$
6	$\frac{15}{2^3} \sqrt{\frac{\pi}{a^7}}$
8	$\frac{105}{2^4} \sqrt{\frac{\pi}{a^9}}$

$$\begin{aligned}
 Z(\beta) &\approx \int_{-\infty}^{\infty} e^{-\beta p^2/2m} dp \int_{-\infty}^{\infty} e^{-\beta a q^2} \left[1 + \underbrace{\beta(bq^3 + cq^4)}_{\substack{\text{odd} \\ \rightarrow 0}} + \frac{\beta^2}{2} \underbrace{(bq^3 + cq^4)^2}_{\substack{Aq^6 + \cancel{Bq^7} + Cq^8 \\ \text{odd}}} \right] dq \cdot \frac{1}{h} \\
 &= \left(\frac{2m\pi}{\beta} \right)^{1/2} \cdot \left[\left(\frac{\pi}{\beta a} \right)^{1/2} + \frac{3}{4} \beta c \left(\frac{\pi}{\beta^5 a^5} \right)^{1/2} + \frac{15}{16} \beta^2 b^2 \left(\frac{\pi}{\beta^7 a^7} \right)^{1/2} + \frac{105}{32} \beta^2 c^2 \left(\frac{\pi}{\beta^9 a^9} \right)^{1/2} \right] \cdot \frac{1}{h} \\
 &= \left(\frac{2m\pi}{\beta} \right)^{1/2} \cdot \left(\frac{\pi}{\beta a} \right)^{1/2} \cdot \frac{1}{h} \left[1 + \frac{3c}{4\beta a^2} + \frac{15b^2}{16\beta a^3} + \frac{105c^2}{32\beta^2 a^4} \right]
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 \ln[Z(\beta)] &= \ln\left(\frac{\pi}{\beta h} \cdot \left(\frac{2m}{a}\right)^{1/2}\right) + \ln\left[1 + \frac{3c}{4\beta a^2} + \frac{15b^2}{16\beta a^3} + \frac{105c^2}{32\beta^2 a^4}\right] \\
 &= \ln\left(\frac{\pi}{h} \left(\frac{2m}{a}\right)^{1/2}\right) - \ln\beta + \frac{3c}{4\beta a^2} + \frac{15b^2}{16\beta a^3} + \frac{105c^2}{32\beta^2 a^4} \quad \text{expansion of } \ln(1+x)
 \end{aligned}$$

We have from 3.5.14

$$\begin{aligned}
 U &= - \left[\frac{\partial}{\partial \beta} \ln(Z(\beta)) \right]_E \\
 &= \frac{1}{\beta} + \frac{3c}{4\beta^2 a^2} + \frac{15b^2}{16\beta^2 a^3} + \frac{105c^2}{16\beta^3 a^4} \\
 &= kT + \frac{3c}{4a^2} k^2 T^2 + \frac{15b^2}{16a^3} k^2 T^2 + \frac{105c^2}{16a^4} k^3 T^3
 \end{aligned}$$

Finally,

$$C = \frac{\partial U}{\partial T} = k + \frac{3ck^2}{2a^2} T + \frac{15b^2 k^2}{8a^3} T + \frac{315c^2 k^3}{16a^4} T^2$$

Leading Contribution

$$\frac{3}{2} k^2 \left[\frac{c}{a^2} + \frac{5b^2}{4a^3} \right] T$$

The mean value $\langle q \rangle$ can be found by

$$\langle q \rangle = \frac{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} q e^{-\beta H} dq dp}{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\beta H} dq dp}$$

$$= \frac{\int_{-\infty}^{\infty} q \exp \{-\beta a q^2\} [1 + \beta(bq^3 + cq^4) + \dots] dq}{\int_{-\infty}^{\infty} \exp \{-\beta a q^2\} dq}$$

finding the first term contribution.

$$\approx \frac{\beta b \int_{-\infty}^{\infty} q^4 e^{-\beta a q^2} dq}{\int_{-\infty}^{\infty} e^{-\beta a q^2} dq}$$

since $\int_{-\infty}^{\infty} q e^{-q^2} \rightarrow 0$

$$= \beta b \frac{\frac{3}{4} \left(\frac{\pi}{\beta^5 a^5} \right)^{1/2}}{\left(\frac{\pi}{\beta a} \right)^{1/2}} = \frac{3b}{4\beta a^2} = \frac{3}{4} \frac{bkT}{a^2} \quad \square$$

4. Now let's look at a quantum-mechanical, one-dimensional, anharmonic oscillator. The energy levels can be approximated as

$$\epsilon_n = \left(n + \frac{1}{2}\right) \hbar\omega - x \left(n + \frac{1}{2}\right)^2 \hbar\omega, \quad n = 0, 1, 2 \dots$$

The parameter defining the anharmonicity is typically $x \ll 1$. Show that to the first order in x and to 4th order in $u = \hbar\omega/kT$, the specific heat of N of such oscillators is given by

$$C = NK \left[\left(1 - \frac{1}{12}u^2 + \frac{1}{240}u^4\right) + 4x \left(\frac{1}{u} + \frac{1}{80}u^3\right) \right], \text{ for high temperatures } (u \ll 1).$$

Goal & Find the specific heat for an anharmonic oscillator with energy levels ϵ_n .

Solution

Since discrete energy levels are given rather than a continuous hamiltonian $H(p, q)$,

We investigate the single-oscillator partition function &

$$\begin{aligned} Z_1(\beta) &= \sum_{n=0}^{\infty} e^{-\beta \epsilon_n} = \sum_{n=0}^{\infty} e^{-\beta \left[(n + \frac{1}{2}) \hbar\omega - x (n + \frac{1}{2})^2 \hbar\omega \right]} \\ &= \sum_{n=0}^{\infty} e^{-\beta (n + \frac{1}{2}) \hbar\omega} \cdot e^{\beta x (n + \frac{1}{2})^2 \hbar\omega} \\ &\approx \sum_{n=0}^{\infty} e^{-\beta (n + \frac{1}{2}) \hbar\omega} \cdot \left[1 + \beta x (n + \frac{1}{2})^2 \hbar\omega \right] \quad x \ll 1 \text{ using expansion of } e^{ax} \\ &= \underbrace{\sum_{n=0}^{\infty} e^{-\beta (n + \frac{1}{2}) \hbar\omega}}_{Z_h(\beta)} + \beta \hbar\omega x \underbrace{\sum_{n=0}^{\infty} (n + \frac{1}{2})^2 e^{-\beta (n + \frac{1}{2}) \hbar\omega}}_{\frac{1}{(\hbar\omega)^2} \frac{\partial^2}{\partial \beta^2} Z_h(\beta)} \end{aligned}$$

Where $Z_h(\beta)$ is the harmonic partition function found in 3.8.14

$$Z_h(\beta) = \left[2 \sinh\left(\frac{1}{2} \beta \hbar\omega\right) \right]^{-1} \quad (3.8.14)$$

$$\begin{aligned} \frac{\partial}{\partial \beta} \left[\frac{\partial}{\partial \beta} Z_h(\beta) \right] &= \frac{\partial}{\partial \beta} \left[-\frac{1}{2} \frac{\frac{1}{2} \hbar\omega \cosh(\frac{1}{2} \beta \hbar\omega)}{\sinh^2(\frac{1}{2} \beta \hbar\omega)} \right] \\ &= \frac{\partial}{\partial \beta} \left[-\frac{1}{4} \hbar\omega \frac{\coth(\frac{1}{2} \beta \hbar\omega)}{\sinh(\frac{1}{2} \beta \hbar\omega)} \right] \\ &= -\frac{1}{4} \hbar\omega \left[\coth(\frac{1}{2} \beta \hbar\omega) \cdot -\frac{1}{2} \hbar\omega \frac{\coth(\frac{1}{2} \beta \hbar\omega)}{\sinh(\frac{1}{2} \beta \hbar\omega)} - \frac{\frac{1}{2} \hbar\omega \operatorname{csch}^2(\frac{1}{2} \beta \hbar\omega)}{\sinh(\frac{1}{2} \beta \hbar\omega)} \right] \\ &= \frac{(\hbar\omega)^2}{8} \sinh(\frac{1}{2} \beta \hbar\omega)^{-1} \left[\coth^2(\frac{1}{2} \beta \hbar\omega) + \operatorname{csch}^2(\frac{1}{2} \beta \hbar\omega) \right] \\ &= \frac{(\hbar\omega)^2}{8} \sinh(\frac{1}{2} \beta \hbar\omega)^{-1} \left[1 + 2 \operatorname{csch}^2(\frac{1}{2} \beta \hbar\omega) \right] \end{aligned}$$

Replacing $U = \beta \hbar \omega$ in $Z_1(\beta)$ with the respective functions $Z_h(\beta)$ & $\partial_\beta^2 Z_h(\beta)$ to get

$$Z_1(u) = \left\{ 2 \sinh \frac{u}{2} \right\}^{-1} + \frac{u\hbar}{8} \left\{ \sinh \frac{u}{2} \right\}^{-1} \left[1 + 2 \operatorname{csch}^2 \left(\frac{u}{2} \right) \right]$$

Taking logarithms to find Energy U .

$$\begin{aligned} \ln [Z_1(u)] &= \ln \left[\left\{ 2 \sinh \frac{u}{2} \right\}^{-1} + \frac{u\hbar}{8} \left\{ \sinh \frac{u}{2} \right\}^{-1} \left[1 + 2 \operatorname{csch}^2 \left(\frac{u}{2} \right) \right] \right] \\ &= \ln \left[\left\{ 2 \sinh \frac{u}{2} \right\}^{-1} \left(1 + \frac{u\hbar}{4} \left[1 + 2 \operatorname{csch}^2 \frac{u}{2} \right] \right) \right] \\ &= \underbrace{-\ln [2 \sinh \frac{u}{2}]}_{Z_h(u)} + \underbrace{\frac{u\hbar}{4} (1 + 2 \operatorname{csch}^2 \frac{u}{2})}_{Z_{ah}(u)} \end{aligned}$$

For the $Z_h(u)$ term, we already derived the heat capacity in p. 67.

Let's proceed to redrive it.

Important Relations:

$$U = \beta \hbar \omega$$

$$U = \frac{\hbar \omega}{kT}$$

$$\frac{\partial}{\partial U} = \frac{1}{\hbar \omega} \cdot \frac{\partial}{\partial \beta}$$

$$\frac{\partial}{\partial U} = \frac{-kT^2}{\hbar \omega} \frac{\partial}{\partial T} = -\frac{\hbar \omega}{k} \frac{1}{U^2} \frac{\partial}{\partial T}$$

$$\begin{aligned} U &= -\frac{\partial}{\partial \beta} Z_h(u) = -\hbar \omega \frac{\partial}{\partial u} Z_h(u) \\ &= \hbar \omega \frac{\partial}{\partial u} \ln [2 \sinh \frac{u}{2}] \\ &= \frac{\hbar \omega \cosh(\frac{1}{2}u)}{2 \sinh(\frac{1}{2}u)} \\ &= \frac{\hbar \omega}{2} \coth(\frac{1}{2}u) \end{aligned}$$

$$\begin{aligned} C &= \frac{\partial U}{\partial T} = -\frac{k}{\hbar \omega} U^2 \frac{\partial}{\partial u} U \\ &= \frac{kU^2}{2} \frac{\partial}{\partial u} \coth(\frac{1}{2}u) \\ &= \frac{kU^2}{4} \operatorname{csch}^2(\frac{1}{2}u) \end{aligned}$$

Replacing $\operatorname{csch}^2(\frac{1}{2}u)$ with its Laurent Expansion. (A.2)

$$\begin{aligned} C &= \frac{kU^2}{4} \left(\frac{4}{U^2} - \frac{1}{3} + \frac{U^2}{60} + O(U^3) \right) \\ &= k \left(1 - \frac{U^2}{12} + \frac{U^4}{240} + O(U^5) \right) \end{aligned}$$

This is the contribution of 1 harmonic particle on the heat capacity C . Multiplying by N yields the total heat capacity first term in question.

To get the second term in x , we write $Z_{ah}(u)$ in its Laurent series expansion & find C .

$$\begin{aligned} Z_{ah}(u) &= \frac{ux}{4} (1 + 2\text{csch}^2 \frac{u}{2}) \\ &= \frac{ux}{4} \left(1 + 2 \left(\frac{4}{u^2} - \frac{1}{3} + \frac{u^2}{60} \right) + O(u^4) \right) \\ &= x \left(\frac{2}{u} + \frac{u}{12} + \frac{u^3}{120} + O(u^5) \right) \end{aligned}$$

$$\begin{aligned} U_{ah} &= -\frac{\partial}{\partial \beta} Z_{ah}(u) = -\hbar\omega \frac{\partial}{\partial u} Z_{ah}(u) \\ &= \hbar\omega x \left(-\frac{2}{u^2} + \frac{1}{12} + \frac{u^2}{40} \right) \end{aligned}$$

$$\begin{aligned} C_{ah} &= \frac{\partial U}{\partial T} = -\frac{k}{\hbar\omega} u^2 \frac{\partial}{\partial u} U_{ah} \\ &= -kx u^2 \left(\frac{4}{u^3} - \frac{u}{20} \right) \\ &= 4kx \left(\frac{1}{u} + \frac{u^3}{80} \right) \end{aligned}$$

This is the contribution to the heat capacity from anharmonicity of a particle. Multiplying by N brings the total heat capacity which is the second term in the question.

$$\begin{aligned} C_{\text{Total}} &= N \left(C_h(u) + C_{ah}(u) \right) \\ &= NK \left[\left(1 - \frac{u^2}{12} + \frac{u^4}{240} \right) + 4x \left(\frac{1}{u} + \frac{u^3}{80} \right) \right] \end{aligned}$$

Appendix

A1 Integral of $\int_{-\infty}^{\infty} x^n e^{-ax^2} dx$.

$$\int_{-\infty}^{\infty} x^n e^{-x^2} dx \quad ?$$

This is a very interesting integral that requires cool techniques to solve.

Integration by parts yields a clean formula. However, since we are in a physics course, why not do Feynman's Technique?

For Feynman's Technique to work, let's solve the generalized problem $\int_{-\infty}^{\infty} x^n e^{-ax^2} dx$

For $n=0$, $\int_{-\infty}^{\infty} e^{-ax^2} dx$ is just a gaussian with variance $\sigma^2 = \frac{1}{2a}$!

The normalization constant is $\frac{1}{\sqrt{2\pi\sigma^2}} = \frac{\sqrt{a}}{\sqrt{\pi}}$. Therefore, $\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$

For $n = \text{odd } \mathbb{N}$, $\int_{-\infty}^{\infty} \underbrace{x^n}_{\text{odd}} \underbrace{e^{-ax^2}}_{\text{even}} dx = 0$ since the integrand is an odd function.

For $n = \text{even } \mathbb{N}$, we can generate the solution by Feynman's Technique.

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\pi} a^{-1/2}$$

$$\frac{d}{da} \int_{-\infty}^{\infty} x^2 e^{-ax^2} dx = -\frac{\sqrt{\pi}}{2} a^{-3/2}$$

$$\frac{d^2}{da^2} \int_{-\infty}^{\infty} x^4 e^{-ax^2} dx = \frac{\sqrt{\pi}}{2} \cdot \frac{3}{2} a^{-5/2}$$

$\vdots$

$$\begin{aligned} \frac{d^n}{da^n} \int_{-\infty}^{\infty} x^{2k} e^{-ax^2} dx &= \sqrt{\pi} \cdot \frac{2k-1}{2} \cdot \frac{2k-3}{2} \cdots \frac{1}{2} a^{-\frac{2k+1}{2}} \\ &= \sqrt{\pi} \frac{(2k-1)!!}{2^k} a^{-\frac{2k+1}{2}} \end{aligned}$$

For $a=1$, we have that $\int_{-\infty}^{\infty} x^{2k} e^{-x^2} dx = \sqrt{\pi} \frac{(2k-1)!!}{2^k}$.

